Karan Gupta

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Experience

Full Stack Developer Jun 2026 – Present
XCEED, NIT Jalandhar
ILEED – Intelligent Learning Engagement and Entity Detection (Node.js, Express, React, Python, MongoDB, FAISS, ONNX)

- Authored **145 PRs (135 merged)** and **560+ commits** for an AI attendance platform serving **1,000+ students** across **30+ cameras**, spanning the Node.js backend, React client, Python ML service, and mobile app.
- Built real-time student identification from live RTSP streams and video uploads with a FAISS vector index and RetinaFace (ONNX), plus automated ERP face-embedding generation and nightly rebuilds.
- Migrated RTSP recording to the Node.js backend with scheduled recordings and FFmpeg audio extraction; added live-preview auto-retry and stale-frame detection for unreliable camera streams.
- Resolved production bottlenecks behind **11.9s** responses and an **11.3%** timeout rate, fixed an unbounded stream memory leak (heap at 91%), and cut ERP list loads from **2.0s to 915ms**.
- Shipped one-click deploys with health checks and automatic rollback, a GPU inventory dashboard with MIG/VRAM telemetry, and uptime alerting; fixed broken access control and data-leak vulnerabilities.

XCEED Learning App – Android & iOS (Capacitor, React, Firebase Cloud Messaging, TanStack Query, GitHub Actions)

- Built the app from scratch with PIN/biometric sign-in, deep links, offline caching, and FCM push notifications.
- Designed a delta OTA update pipeline on GitHub Actions that ships only changed files (a 3-file release: **0.41 MB instead of 21.6 MB**), and automated web-to-app code sync with custom merge drivers.

Projects

EventFlow – Distributed Workflow Orchestration Engine
Tech: FastAPI, Python, PostgreSQL, Redis Streams, gRPC, Docker, WebSockets
GitHub: github.com/guptakaran20/EventFlow | Live: goeventflow.vercel.app

- Built a workflow engine with DAG routing, distributed Redis Streams workers, retries, and a Dead Letter Queue.
- Added fault tolerance with heartbeats, stuck-job recovery (`XPENDING/XCLAIM`), and live WebSocket updates.

CodeArena – Real-Time Coding Battle Platform
Tech: Next.js, Node.js, Express, MongoDB, Redis, Socket.IO, Docker, AWS EC2
GitHub: github.com/guptakaran20/CodeBattle | Live: codearenabattle.vercel.app

- Built a real-time multiplayer coding platform with battle rooms, Redis matchmaking, tournaments, leaderboards.
- Built JWT/OAuth REST APIs and Piston code execution; deployed on AWS EC2 + Nginx with Prometheus metrics.

ImportlyAI – AI-Powered CSV Import Platform
Tech: Next.js, TypeScript, Node.js, Express.js, Gemini API, PapaParse
GitHub: github.com/guptakaran20/ImportlyAI | Live: importlyai.vercel.app

- Engineered a multi-stage Gemini pipeline mapping arbitrary CSVs to a standard CRM schema, with SSE progress.
- Developed a benchmark suite with 180+ tests and optimized batched inference for large, inconsistent datasets.

SponsorGrid – Sponsorship Management Platform
Tech: Next.js, TypeScript, PostgreSQL, Prisma ORM, Zod, Cloudinary
GitHub: github.com/guptakaran20/Sponsorship | Live: trysponsorgrid.vercel.app

- Built a SaaS platform for sponsor and funding workflows with Google OAuth, CSRF-protected cookie auth, and Zod.

Technical Skills

Languages: C++, TypeScript, JavaScript, Python, SQL
Frontend: React.js, Next.js, Tailwind CSS, TanStack Query, HTML5, CSS3
Backend: Node.js, Express.js, FastAPI, REST APIs, gRPC, WebSockets, Socket.IO, JWT, OAuth
Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM
AI & ML: FAISS, ONNX Runtime, Face Recognition, Gemini API
DevOps & Tools: Docker, AWS EC2, Nginx, GitHub Actions, CI/CD, PM2, Prometheus metrics, Git, Vercel
Mobile: Capacitor, Android, iOS, Firebase Cloud Messaging, OTA Updates

Open Source

- **GSSoC'26:** Merged an advanced-level PR hardening auth validation and API responses in a MERN leave system.
- **NITJ Aarogya Club:** 9 merged PRs building a real-time quiz platform (WebSockets, rotating QR tokens, anti-cheat).

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar
B.Tech – Instrumentation and Control Engineering | **CGPA:** 8.49 2025 – 2029

Achievements

- **LeetCode Knight:** 1870 rating (**top 5.7% globally**), 613 problems solved (88 Hard), best contest rank 2,171.
- **1,400+ GitHub contributions** in the past year; LeetCode 200 Days Badge 2026 for consistent daily problem solving.